ECON 0150 | FINAL PROJECT

NFL Scoring & Game-Time Temperature

THE QUESTION

Does game-time temperature affect total scoring in NFL games?

THE DATA

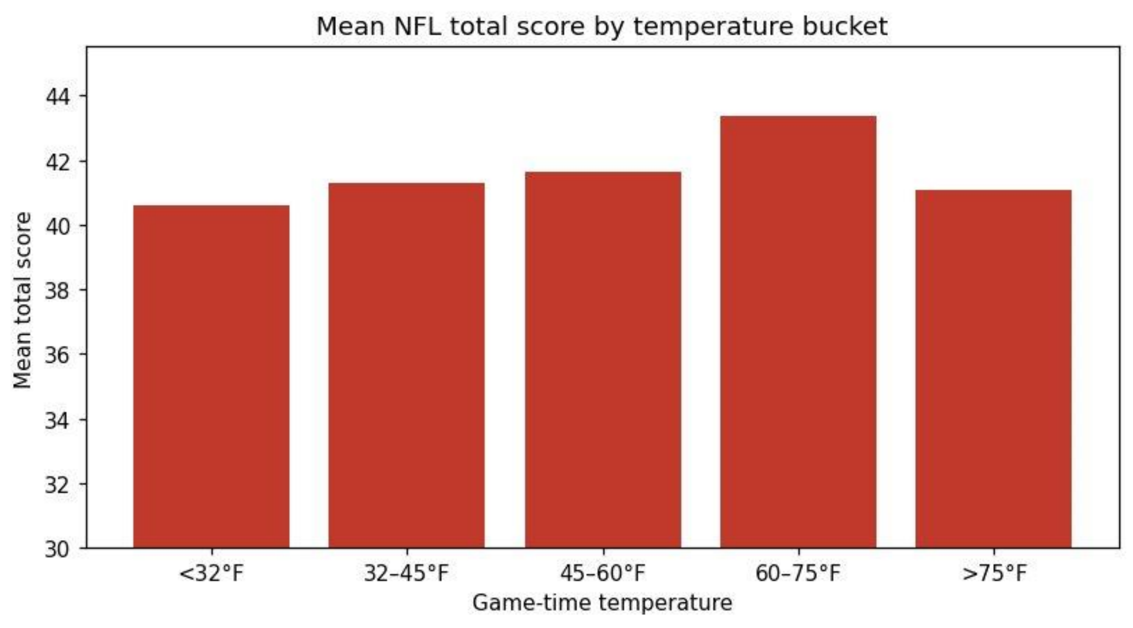
12,844 outdoor NFL games | 1966–2025

Source: Kaggle (NFL Scores & Betting Data)

Mean total score: **42.3 pts (SD 14.4)**
Mean temperature: **59 °F (range –9 to 97)**
Mean wind: **7.5 mph (range 0 to 40)**

What the raw data show

Mean total score by temperature bucket



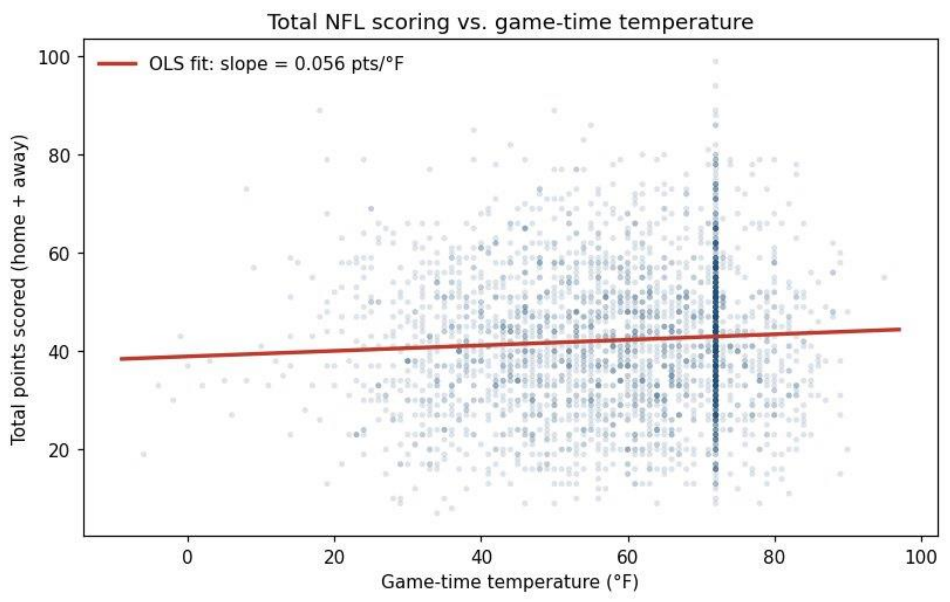
Warmer games score more. The 60–75 °F bucket averages 43.4 pts vs. 40.6 pts for the coldest games.

The temperature effect goes away once we add wind

MODEL (OLS)

Part 4: $total\_score = \beta_0 + \beta_1 \cdot temperature + \varepsilon$

Part 5: $total\_score = \beta_0 + \beta_1 \cdot temperature + \beta_2 \cdot wind + \varepsilon$ ← adds wind as control



Residual plot shows linearity holds and spread is roughly constant (no obvious heteroskedasticity)

+0.056 *pts per °F*

Part 4: just temperature

Significant ($p < 0.001$) but tiny: a 30 °F swing predicts only 1.7 more pts. $R^2 = 0.004$.

−0.005 *pts per °F*

Part 5: adding wind

Temperature falls to about zero and is no longer significant ($p = 0.60$).

−0.38 *pts per mph*

Wind is the real driver

10 mph more wind ≈ 3.8 fewer points per game ($p < 0.001$).

Bottom line: Cold games score less because they are windy. Wind matters, temperature does not.